INSIGHT REPORT

Coffee Shop Sales Analysis Dashboard

Project Insight Report | Microsoft Power BI

Internship	Data Analytics Internship
Organisation	Labmentix
Domain	Data Analytics and Business Intelligence
Tool	Microsoft Power BI Desktop
Dataset	Coffee Shop Sales.xlsx (worksheet: Transactions)
Analysis period	1 January 2023 – 30 June 2023
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1. Executive Summary

This project transforms 149,116 coffee shop transaction records into an interactive Power BI dashboard that summarises sales performance across products, stores, months and hours of the day.

The key findings are:

- **Total revenue** for the six-month period was approximately **698.81K**, from **149,116 transactions** and **214,470 units** sold.
- **Coffee** is the leading category (269.95K, about 38.6% of revenue), followed by **Tea** (196.41K, about 28.1%). Together they account for roughly two-thirds of all revenue.
- Revenue is **evenly split across the three stores**: Hell's Kitchen (33.84%), Astoria (33.23%) and Lower Manhattan (32.92%).
- Monthly revenue **more than doubled** between January (81.68K) and June (166.49K), with a small dip in February.
- Sales peak during the **morning and late-morning hours** and decline later in the day.

The dashboard gives decision-makers a consolidated, filterable view of these indicators and demonstrates an end-to-end workflow: data preparation, data modelling, DAX calculations and visual design.

2. Introduction

The Coffee Shop Sales Analysis Dashboard was developed as part of the Labmentix Data Analytics Internship. Its purpose is to convert raw transactional data into business insight through data preparation, calculated fields, data modelling and interactive visualisation.

The dashboard allows users to explore performance by store location, product category, month and date range, supported by KPI cards and five analytical charts.

3. Project Objectives

1. Understand and assess the quality of the coffee shop transaction dataset.
2. Prepare the data for analysis, including calculated fields and a date table.
3. Calculate total revenue, total transactions, total quantity sold, average transaction value and average daily sales.
4. Compare revenue across store locations, product categories and product types.
5. Analyse monthly revenue trends and hourly sales patterns.
6. Build an interactive Power BI dashboard with slicers.
7. Communicate findings as clear, actionable business insights.

4. Dataset Description

Item	Detail
File / worksheet	Coffee Shop Sales.xlsx / Transactions
Records	149,116
Columns	11
Period	1 January 2023 – 30 June 2023 (6 months)
Store locations	3 (Hell's Kitchen, Astoria, Lower Manhattan)
Missing values	None identified in initial inspection
Duplicate rows	None identified in initial inspection

4.1 Data Dictionary

Column	Description	Data type
transaction_id	Unique identifier for each transaction	Whole Number
transaction_date	Date of the transaction	Date
transaction_time	Time of the transaction	Time
transaction_qty	Quantity of products purchased	Whole Number
store_id	Unique identifier of the store	Whole Number

Column	Description	Data type
store_location	Location of the store	Text
product_id	Unique identifier of the product	Whole Number
unit_price	Price of one unit	Decimal Number
product_category	Broad product group	Text
product_type	Specific product type	Text
product_detail	Detailed product name	Text

5. Methodology

The project followed a structured analytics workflow:

1. **Data collection and understanding:** reviewed the structure, columns and period covered by the dataset.
2. **Data import:** loaded the Transactions worksheet into Power BI Desktop.
3. **Data quality checks:** inspected column names, record counts, null values, duplicates and data types.
4. **Data preparation:** assigned correct data types and created calculated columns.
5. **Data modelling:** created a Date Table and linked it to the Transactions table.
6. **DAX development:** wrote measures for the dashboard KPIs.
7. **Visualisation:** built KPI cards, charts and slicers in Report View.
8. **Validation:** tested slicer interactions and checked figures.
9. **Insight generation:** interpreted the results and documented findings.

6. Data Preparation and Modelling

6.1 Calculated Columns in the Transactions Table

Column	DAX	Purpose
Sales	Transactions[transaction_qty] * Transactions[unit_price]	Revenue value for each transaction record
Transaction Hour	HOUR(Transactions[transaction_time])	Extracts the hour for time-of-day analysis

6.2 Date Table

A separate Date Table was created using CALENDAR(MIN(Transactions[transaction_date]), MAX(Transactions[transaction_date])), which produces a continuous calendar covering the analysis period. Four calculated columns were added:

Column	DAX	Purpose
Month Name	FORMAT('Date Table'[Date], "MMMM")	Full month name
Month Number	MONTH('Date Table'[Date])	Numeric month, used to sort Month Name chronologically
Year	YEAR('Date Table'[Date])	Calendar year
Year-Month	FORMAT('Date Table'[Date], "YYYY-MM")	Year-and-month label

Month Name was sorted by Month Number so that months display in the order January → June instead of alphabetically.

6.3 Relationship and Model Design

Property	Configuration
Primary (one) side	Date Table[Date]
Related (many) side	Transactions[transaction_date]
Cardinality	One-to-many (1:*)
Cross-filter direction	Single
Status	Active

The Date Table was also marked as the official date table in Power BI. This supports date filtering, monthly comparison and future time-intelligence calculations.

7. DAX Measures

Measure	DAX	Result
Total Revenue	SUM(Transactions[Sales])	698,812.33
Total Quantity	SUM(Transactions[transaction_qty])	214,470 units
Total Transactions	DISTINCTCOUNT(Transactions[transaction_id])	149,116
Average Transaction Value	DIVIDE([Total Revenue], [Total Transactions])	4.69
Average Daily Sales	AVERAGEX(VALUES('Date Table'[Date]), [Total Revenue])	3.86K per day

DIVIDE is used instead of the / operator because it handles division by zero safely. DISTINCTCOUNT counts unique transaction IDs rather than rows. Revenue by store, category and product type reuses **Total Revenue** with the relevant field supplied by each chart's filter context, so no separate measures were needed.

8. Dashboard Design

8.1 Layout

The dashboard contains a title and subtitle, five KPI cards, five analytical charts and four interactive slicers (store location, product category, month and date range).

8.2 KPI Cards

KPI	Measure	Value
Total Revenue	Total Revenue	698.81K
Cups & Items Sold	Total Quantity	214K
Total Orders	Total Transactions	149K
Average Transaction Value	Average Transaction Value	4.69
Average Daily Sales	Average Daily Sales	3.86K

8.3 Visualisations

Visual	Type	Fields	Purpose
Revenue by Product Category	Horizontal clustered bar	product_category, Total Revenue	Compare category contribution
Monthly Revenue	Clustered column	Month Name, Total Revenue	Show the monthly trend
Revenue by Product Type	Horizontal clustered bar	product_type, Total Revenue	Detailed product-level comparison
Revenue by Store Location	Donut	store_location, Total Revenue	Show each store's share
Revenue by Hour of Day	Line / area	Transaction Hour, Total Revenue	Identify peak trading hours

8.4 Visual Theme

A coffee-inspired palette was applied consistently across the report.

Colour	Hex	Use
Dark Mocha	#4B2E24	Titles and emphasis
Coffee Brown	#754C38	Secondary elements
Caramel	#A77959	Highlights
Latte Beige	#D9B99B	Supporting backgrounds
Cream	#F5EBDD	Light backgrounds

9. Analysis and Key Findings

9.1 Overall Performance

The business generated about 698.81K in revenue from 149,116 transactions over six months. The average transaction was worth 4.69, with about 1.44 units per transaction, and average daily revenue was about 3.86K.

9.2 Product Category Performance

Category	Revenue	Share of total
Coffee	269.95K	38.6%
Tea	196.41K	28.1%
Bakery	82.32K	11.8%
Drinking Chocolate	72.42K	10.4%
Coffee Beans	40.09K	5.7%
Branded	13.61K	1.9%
Loose Tea	11.21K	1.6%
Flavours	8.41K	1.2%
Packaged Chocolate	4.41K	0.6%

Coffee and Tea together contribute about 66.7% of revenue, which makes them the core of the sales mix. Bakery and Drinking Chocolate form a secondary tier, and the remaining five categories together contribute about 11% of revenue.

9.3 Monthly Revenue Trend

Month	Revenue	Change vs previous month
January	81.68K	–
February	76.15K	−6.8%
March	98.83K	+29.8%
April	118.94K	+20.3%
May	156.73K	+31.8%
June	166.49K	+6.2%

February was the lowest month and June the highest. Revenue grew in every month from March onwards, and June revenue was roughly double January's. Growth slowed in June relative to the preceding months.

9.4 Store Performance

Store	Revenue	Share
Hell's Kitchen	236.51K	33.84%
Astoria	232.24K	33.23%
Lower Manhattan	230.06K	32.92%

The gap between the highest and lowest store is about 6.45K (under 1% of total revenue), so performance is well balanced across locations.

9.5 Product-Type Performance

The product-type chart breaks the category totals down into individual product types, showing which specific products drive revenue within Coffee, Tea and Bakery and which contribute little. It supports product-level inventory and promotion decisions.

9.6 Hourly Sales Pattern

Revenue rises through the morning, reaches its highest levels in the late-morning period and generally declines later in the day.

10. Business Recommendations

The following recommendations are based on the observed patterns. The data does not establish the causes behind them, so they should be treated as starting points for further review.

1. **Prioritise Coffee and Tea in inventory and promotion planning.** They generate about two-thirds of revenue, so stock-outs here have the largest impact.
2. **Align staffing and preparation with the morning and late-morning peak.** Ensuring adequate staff and prepared stock in these hours could protect the highest-revenue period.
3. **Review low-contribution categories.** Branded, Loose Tea, Flavours and Packaged Chocolate each contribute under 2% of revenue. Management could assess whether they warrant their shelf space or need targeted promotion.
4. **Share practices across stores.** Because the three stores perform almost identically, small differences, such as Hell's Kitchen's slight lead, are better investigated through operational factors than assumed to be structural.
5. **Investigate the growth drivers and the February dip.** Management should check whether promotions, seasonality, new products or opening-hour changes explain the pattern, then plan to sustain or repeat it.
6. **Use product-type data for range decisions.** The long tail of low-revenue product types is a candidate for range rationalisation.

11. Limitations

1. **Short period:** six months of data limits trend and seasonality conclusions.
2. **No cost data:** profit and margin cannot be calculated.

3. **No customer identifiers:** segmentation, retention and repeat-purchase analysis are not possible.
4. **Limited context:** revenue patterns alone cannot explain why a month, store or product performed as it did.
5. **Currency not specified:** all values are in the dataset's price units.
6. **No external factors:** promotions, holidays, weather and similar influences are not captured.

12. Future Scope

1. Add profit and margin analysis using cost data.
2. Extend the dataset to additional months or years and add seasonality analysis.
3. Add month-on-month growth measures using the Date Table.
4. Introduce sales forecasting.
5. Add drill-through pages for individual stores and products.
6. Add KPIs such as revenue growth and units per order.
7. Introduce customer segmentation if customer-level data becomes available.

Proposed time-intelligence measures:

```
Previous Month Sales =
CALCULATE([Total Revenue], DATEADD('Date Table'[Date], -1, MONTH))

MoM Revenue Change =
[Total Revenue] - [Previous Month Sales]

MoM Growth % =
DIVIDE([MoM Revenue Change], [Previous Month Sales])
```

13. Skills and Learning Outcomes

Through this project the following skills were developed:

- Importing Excel data into Power BI and validating data quality and types.
- Creating calculated columns, measures and a dedicated Date Table in DAX.
- Designing a data model with a one-to-many relationship.
- Selecting suitable visuals for different comparisons.
- Building interactive dashboards with slicers and a consistent visual theme.
- Interpreting results and communicating them as business insights.

14. Conclusion

The Coffee Shop Sales Analysis Dashboard shows how raw transactional data can be turned into clear business insight using Power BI. The analysis found that Coffee and Tea drive most revenue, the three stores perform almost equally, revenue grew strongly through the first half of 2023, and sales concentrate in the morning and late-morning hours.

The project provided practical experience across the full analytics workflow, from data preparation and modelling to DAX, visualisation and insight generation, and it forms a foundation for deeper profitability, forecasting and customer analysis if additional data becomes available.

Appendix A – Summary of Key Metrics

Metric	Value
Total Revenue	698,812.33
Total Transactions	149,116
Total Quantity Sold	214,470
Average Transaction Value	4.69
Average Daily Sales	3.86K
Highest Revenue Month	June (166.49K)
Lowest Revenue Month	February (76.15K)
Highest Revenue Category	Coffee (269.95K)
Highest Revenue Store	Hell's Kitchen (236.51K)

Appendix B – Tools and Technologies

Tool	Purpose
Microsoft Excel	Source dataset and initial inspection
Power BI Desktop (Data, Model and Report views)	Data preparation, modelling and dashboard development
DAX	Calculated columns, tables and measures
Card, bar, column, donut and line/area visuals	KPI display and comparative charts
Slicers	Interactive filtering by store, category, month and date

Note: All revenue figures are derived from the supplied dataset and expressed in its price units. Percentage shares and month-on-month changes in Sections 9 and 10 were calculated from the dashboard figures above.